loki23
smart_ptr
FileDescriptorStorage T
-handle_ : int = -1
+get() : int
+set(h : int)
+null_handle() : int
+is_valid(h : int) : bool
+destroy(h : int)
SocketStorage T
-handle_ : int = -1
+get() : int
+set(h : int)
+null_handle() : int
+is_valid(h : int) : bool
+destroy(h : int)
DefaultStorage T
-ptr_ : T* = nullptr
+get() : T*
+set(h : T*)
+null_handle() : T*
+is_valid(h : T*) : bool
+destroy(h : T*)
LockedStorage T
-ptr_ : T*
-mutex_ : std::mutex
+get() : T*
+set(h : T*)
+null_handle() : T*
+is_valid(h : T*) : bool
+destroy(h : T*)
RefCountedMT
+acquire(h)
+clone(h) : handle_type
+release(h) : bool
+use_count(h) : int
Copy
+acquire(h)
+clone(h) : handle_type
+release(h) : bool
+use_count(h) : int
DeepCopy
+clone(h) : handle_type
+release(h) : bool
SharedOwnership
+acquire(h)
+clone(h) : handle_type
+release(h) : bool
+use_count(h) : int
RefCounted
+acquire(h)
+clone(h) : handle_type
+release(h) : bool
+use_count(h) : int
RefLinked
-list_ : std::list<RefLinked*>
+acquire(h)
+clone(h) : handle_type
+release(h) : bool
NoCopy
+clone(h) = delete
+release(h) : bool
SmartPtr Tval, OwnPolicy, CheckPolicy, StorePolicy
-storage_ : StorePolicy
-ownership_ : OwnPolicy
+SmartPtr()
+SmartPtr(h : handle_type)
+SmartPtr(rhs : SmartPtr const&)
+SmartPtr(rhs : SmartPtr&&)
+~SmartPtr()
+operator=(rhs : SmartPtr const&) : SmartPtr&
+operator=(rhs : SmartPtr&&) : SmartPtr&
+get() : handle_type
+valid() : bool
+use_count() : int
+operator*() : Tval&
+operator->() : Tval*
-clean()
UniqueOwnership
+acquire(h)
+release(h) : bool
+use_count(h) : int
IntrusiveOwnership
+acquire(h)
+release(h) : bool
+use_count(h) : int
ExceptionCheck
+check(valid : bool)
HaltCheck
+check(valid : bool)
NoCheck
+check(valid : bool)
AssertCheck
+check(valid : bool)
AllowConversion
+implicit_convert() : bool
DisallowConversion
+implicit_convert() : bool
StoragePolicy T
+get() : handle_type
+set(h : handle_type)
+null_handle() : handle_type
+is_valid(h : handle_type) : bool
+destroy(h : handle_type)
AtomicRefCountBase
-count_ : std::atomic<std::size_t>*
+alloc_count()
+inc_count()
+dec_count() : bool
+use_count() : std::size_t
Clonable T
+clone(h : handle_type) : handle_type
RefCountBase
-count_ : std::size_t*
+RefCountBase(nullptr_t)
+RefCountBase(rhs : RefCountBase const&)
+RefCountBase(rhs : RefCountBase&&)
+alloc_count()
+inc_count()
+dec_count() : bool
+use_count() : std::size_t
OwnershipPolicy T
+acquire(h : handle_type)
+release(h : handle_type) : bool
CheckingPolicy
+check(valid : bool)
ConversionPolicy
+implicit_convert() : bool
singleton
CreateUsingMalloc T
+create() : T*
+destroy(p : T*)
CreateStatic T
+create() : T*
+destroy(p : T*)
CreateUsingNew T
+create() : T*
+destroy(p : T*)
ConcreteLifetimeTracker T, Destroyer
-tracked_ : T*
-destroyer_ : Destroyer
+~ConcreteLifetimeTracker()
SingletonHolder T, CreationPol, LifetimePol
-pInstance_ : std::atomic<T*>
-destroyed_ : bool
-mutex_ : std::mutex
+instance() : T&
-destroy_singleton()
DefaultLifetime T
+schedule_destruction(p : T*, fn : void(*)())
+on_dead_reference()
PhoenixSingleton T
-destroyed_once_ : bool
+schedule_destruction(p : T*, fn : void(*)())
+on_dead_reference()
SingletonWithLongevity T
+schedule_destruction(p : T*, fn : void(*)())
+on_dead_reference()
NoDestroy T
+schedule_destruction(p : T*, fn : void(*)())
+on_dead_reference()
CreationPolicy T
+create() : T*
+destroy(p : T*)
LifetimeTracker
#longevity_ : unsigned int
+{virtual} ~LifetimeTracker()
+compare(lhs, rhs) : bool
LifetimePolicy T
+schedule_destruction(p : T*, fn : void(*)())
+on_dead_reference()
type_list
Metafunctions: length, type_at, index_of, contains, push_front, push_back, append, erase, erase_all, no_duplicates, replace, replace_all, reverse, most_derived, derived_to_front, apply, filter, for_each
type_list Ts...
di
Container Ts...
+resolve(From<X>) : X&
+resolve(a) : a&&
+bind<T>(args...)
+get<T>() : T&
uses
From X
uses
hierarchy
GenLinearHierarchy Unit, T
Field<T>(obj) -> Unit<T>&
Dep T
-holder : Holder<T>&
LazyHolder T
-creator : std::function<void(T*)>
+bind(args...)
+get() : T& override
ImmediateHolder T
-creator : std::function<void(T*)>
+bind(args...)
DefaultHolder T
-value : T
+get() : T&
+mem_location() : void*
GenScatterHierarchy Unit, T
Field<T>(obj) -> Unit<T>&
Holder T
#initialized : bool
#union value : T
+{virtual} ~Holder()
+{virtual} get() : T&
+mem_location() : void*
+is_initialized() : bool
factory
HierarchyFactory Base, Types, CreationPolicy
NewCreation
+create<Base, Derived>() : SmartPtr<Base, NoCopy>
FactoryMethodCreation
+create<Base, Derived>() : SmartPtr<Base, NoCopy>
Factory Base, Types, CreationPolicy
-m_registry : std::unordered_map<string_view, Creator>
+Factory()
+create(name : string_view) : SmartPtr<Base, NoCopy>
+contains(name : string_view) : bool
FactoryCreationPolicy Base, Derived
+create() : SmartPtr<Base, NoCopy>
multimethods
DispatchPolicy
FIRST_MATCH
ALL_MATCHES
LogNoMatchPolicy
+no_match()
StaticDispatcher DispatchPol, NoMatchPol, BaseLhs, BaseRhs, Ts...
+dispatch(lhs : BaseLhs&, rhs : BaseRhs&, functors...)
NoOpNoMatchPolicy
+no_match()
ExceptionNoMatchPolicy
+no_match()
DynamicDispatcher DispatchPol, NoMatchPol, Args...
-callbacks_ : unordered_map / unordered_multimap
+add<ActualArgs...>(callback : std::function<void(ActualArgs...)>)
+dispatch(args : Args...)
NoMatchPolicy
+no_match()
TypeIndexArrayHash size
+operator()(arr : std::array<type_index, size>) : std::size_t